Fiber Optics - Chromatic dispersion report

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I. Introduction

Modal dispersion is one of the strongest dispersion effects that may take place along an optical fiber. The obvious solution to prevent it is to use single-mode fibers, namely fibers characterized by a shorter radius and/or a smaller numerical aperture.

However, once modal dispersion has been taken care of, other undesired effects become more visible. Among them, chromatic dispersion is the most important one. It is a second-order effect which causes the variation of the group velocity with the wavelength of the electromagnetic radiation, and it is usually characterized by either the coefficient of chromatic dispersion per unit length,

$$\beta_2 = \frac{d^2\beta}{d\omega^2} = \frac{d}{d\omega}\frac{1}{v_g} = -\frac{1}{v_g^2}\frac{dv_g}{d\omega}, \tag{1}$$

or by the so-called D-parameter, defined as

$$D = \frac{d}{d\lambda}\frac{1}{v_g}. \tag{2}$$

Chromatic dispersion is caused by three main phenomena: material, guide and profile dispersion. In terms of the D-parameter, these contributors have opposite signs, and for a typical single-mode, step-index fiber they cancel one another at about 1300 nm. However, it is possible to shift the zero-dispersion wavelength by modifying the refractive index profile, and thus tuning guide dispersion, to the minimum-loss window, at about 1550 nm. Such fibers are called dispersion-shifted fibers (DSF), and they have been replaced by non-zero dispersion-shifted fibers (NZDSF), which are less susceptible to nonlinear effects.

The total chromatic dispersion along an optical fiber of length L can be described by

$$DL = \frac{d\tau_g}{d\lambda}, \tag{3}$$

where τ_g is the group delay, namely the propagation time of a signal along the fiber. This approach lets us easily calculate chromatic dispersion by measuring the group delay at different optical wavelengths.

The purpose of this laboratory report is to show the obtained results after measuring chromatic dispersion of a given, unknown optical fiber and, according to the obtained results, hypothesize its classification. In Section II, a brief illustration of the devices used in the experiment and a summary of the procedure is given. Section III contains all the details of our approach towards the analysis of the data. Finally, Section IV gives a summary of our findings and conclusions from this experiment.

II. Experimental setup

As previously mentioned, in this laboratory experience chromatic dispersion was estimated by measuring the group delay. A representation of the setup is illustrated in Figure 1.

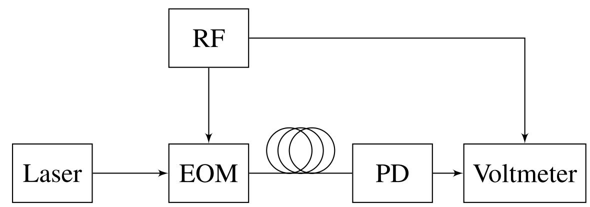

Fig. 1: Experimental setup.

Instead of measuring directly the group delay at different wavelengths, a radio-frequency (RF) generator was exploited to generate a sinusoidal signal at $f_0 = 940$ MHz. This generator has two different outputs; the first one is connected to an electro-optic modulator (EOM), and is used to modulate monochromatic light produced by an external-cavity laser, namely a laser with adjustable wavelength, and modified by a polarization controller.

The modulated optical signal goes through an optical fiber of unknown properties. Since the measures in this experiment are quite sensitive to fluctuations, especially those caused by temperature, the fiber in question was inserted into a box full of foam, in order to decrease the fluctuation rate.

The signal from the fiber is received by a photodiode, which converts it back to an electric signal and sends it to a vector voltmeter, which measures both voltage and phase of two input signals of equal frequency. As a matter of fact, this device also receives the original sinusoidal signal, thanks to a direct link between it and the RF generator, and is able to measure the phase difference $\Delta\phi$ between the two signals. Therefore, the group delay at a particular optical wavelength is given by

$$\tau_g = \frac{\Delta\phi}{2\pi f_0}. \tag{4}$$

The group delay itself does not give particularly useful information about chromatic dispersion. In fact, it is unknown whether the signal has been delayed by more than one period, as well as which signal arrives first at the voltmeter. Fortunately, we are only interested in the derivative of this quantity with respect to wavelength. This is the reason why a tunable laser is used: by taking several measurements at closely-spaced wavelengths, chromatic dispersion can be estimated as given in (3).

In this case, two sets of fifty measurements each were taken, and both are characterized by the same wavelength range $\Delta\lambda$ and wavelength step $d\lambda$, which are shown in Table I.

Details on the analysis approach and the results are given in the following section.

Parameter	Symbol	Value
Refractive index of the fiber core	n_{co}	1.47
Modulating frequency	f_0	940 [MHz]
Wavelength range	$\Delta\lambda$	1560 $\div$ 1610 [nm]
Wavelength step	$d\lambda$	1 [nm]
Fiber length	L	20.7 [km]

TABLE I: Experimental parameters.

III. Results

A. Chromatic Dispersion

As already mentioned in Section II, our measurements were heavily affected by unwanted effects from both the fiber or the laser progressively heating. In particular the laser cavity, which is made of metal, is able to change its volume due to a large expansion coefficient.

The empirical result is that the phase shift for a given wavelength is not a constant value, but it slightly changes over time due to the mentioned effect. Therefore, we had to be quick while observing the readings of the instruments.

Of course, sometimes we were forced to stop and repeat the last measurement, in order to calculate an offset (error) and correct all the following recorded values. These pauses were caused by the rearrangement of the scale value in the vector voltmeter. In fact, the starting scale and the angular offset of the phase detector are not suitable for the whole range of measures (we continue to increase wavelength by $1nm$ steps, according to Table I); indeed, we had to change and configure them from time to time.

For each wavelength we recorded the corresponding phase shift in degrees (fifty values in total), which are represented in Figure 2. In order to underline the effects of source heating, we made two different data collections (with the same setup), the first is the blue one and the second is the orange one.

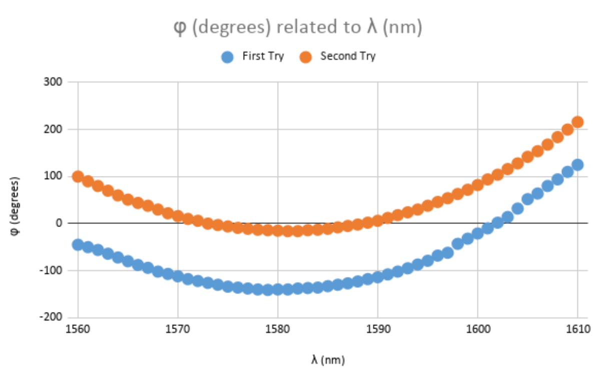

Fig. 2: Phase shift against the wavelength.

Once the phase shifts were collected, we were able to find out the D-parameter in order to evaluate the fiber performance.

First, we computed the group delay from the phase shift by applying (4) for each wavelength. Second, we calculated the differential quantities $d\tau_g$ by subtracting adjacent group delays. Finally, the D-factor is obtained from (3) after dividing for the fiber length L and it is expressed in ps/nm-km.

The relationship between wavelength and chromatic dispersion (D-factor) is then reported in Figure 3.

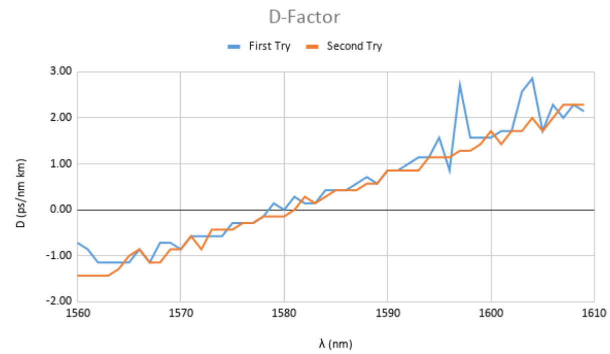

Fig. 3: D-parameter of the fiber against the wavelength.

By looking at the previous graphs, it should be clear that the discontinuities in the blue lines are caused by a temperature increase; on the other hand, once the setup was heated up (after 20 minute from the first data collection), the orange values appear more stable. For this fiber, the wavelength $\lambda \approx$1580 nm is the one where the contributions of chromatic dispersion cancel each other out.

Some considerations can also be made about the chromatic dispersion slope, namely the dependency of chromatic dispersion itself on frequency. Similarly to chromatic dispersion, this quantity can be described with either the third derivative of the propagation constant with respect to angular frequency or, more simply, with the derivative of the D-parameter itself with respect to wavelength:

$$S = \frac{dD}{d\lambda} = \frac{d^2}{d\lambda^2}\frac{1}{v_g}. \tag{5}$$

By considering the latter definition, the dispersion slope S is simply given by the slope of the wavelength-spectrum of D, thus justifying its name. An interpolation of data from Figure 3 gives $S = 0.07$ ps/nm^2-km, which appears to be very close to the typical values of such parameter for silica optical fibers in the third window.

B. Further Considerations on Phase Shift

A remarkable shift in the phase measures is noticeable between the first and the second set of measures in Fig. 2. Even if this is not affecting at all the overall D-parameter response, because of the fact that this value is proportional to the derivative of this measure, it is interesting to further understand the causes of this deviation.

First of all, we point out that the group delay, measured at a given wavelength, must be the same between the two measurement sets, as we are excluding major changes in the properties of the fiber (e.g. the fiber heating up as a consequence of a very high injected power). Such changes would invalidate completely the experiment. Having said that, the group delay is measured indirectly by using the phase difference between the signals arriving at the vector voltmeter.

We also assume that the total phase difference corresponds to the phase accumulated by the wave travelling through the fiber, as the propagation in the direct path from the RF generator to the voltmeter gives no phase retardation in a first approximation.

Therefore, the actual phase difference must be the phase in the voltmeter readout (plotted in Fig. 2) *added* with a *multiple of* $360°$ in order to have physical sense: since the optical waveguide is driven with a wavelength which is far from medium resonances, the group velocity must be less than the speed of light. Hence, the minimum allowable group delay is $\tau_g > L/c = 69$ µs, where $L = 20.7$ km, and therefore the minimum phase difference, given a modulation frequency of 940 MHz, is $\Delta\phi > 23.35° \cdot 10^6$.

These considerations are useful to test the following hypothesis: if $\tau_g = \Delta\phi/(2\pi f_0)$ must be constant, but the same phase measurement at a different time shows a separation of $\delta\phi \approx 150°$, we may suspect that the modulation frequency has undergone a variation of Δf. If the frequency accuracy of the RF generator (which is given by the internal reference oscillator) is not high enough, and it drifted between one measurement set and the other, this could be the source of the shift in the phase measures. If we assume a (minimum) phase difference equal to the one of a modulating wave travelling at the speed of light, the constraint of a constant group delay in between the two set of measures, called here A and B is

$$\tau_g = \frac{\Delta\phi_A}{2\pi f_A} = \frac{\Delta\phi_B}{2\pi f_B},$$

and we can deduce the minimum relative frequency shift to have an effect as the one registered

$$\begin{aligned}\frac{\Delta f}{f_A} &= \frac{\Delta\phi_B}{\Delta\phi_A} - 1 \\ &= \frac{\delta\phi}{\Delta\phi_A} \approx 6 \text{ ppm}.\end{aligned}$$

The RF generator used in the experiment is an Agilent 8648B, from the specifications (shown in Figure 4) we find that the internal reference oscillator is specified to be stable within a few ppm, after a warm-up of 1 hour.

	Standard timebase (typical)	**High stability timebase (Opt 1E5)**
Aging	<±2 ppm/year	<±0.1 ppm/year[2] <±0.0005 ppm/day[2]
Temperature	<±1 ppm	<±0.01 ppm[3] (typical)
Line Voltage[4]	<±0.5 ppm	<±0.1 ppm (typical)

Fig. 4: Stability specification of Agilent RF generator.

However, the phase used for the calculation can eventually be much higher depending on the actual group velocity (we only calculated a lower bound on phase), leading to a smaller frequency drift. We have shown in this way how a modulation frequency drift of a few parts-per-million can eventually affect the phase measure with an important shift.

As regards the external cavity laser, which is an Anritsu Tunics Plus, its wavelength is specified to be stable within ± 5 pm after a warm-up of 1 hour, and accurate within ± 40 pm, so the collected data sets show no shift in the horizontal axis.

Wavelength stability [2]	±5 pm / h (±3 pm / h typical and ±5 pm / 24h typical)

Fig. 5: Stability specification of Anritsu ECL.

In conclusion, although the computation of the dispersion parameter itself can be affected by other slight instabilities in the test setup, as can be seen in the first set of measurements of Figure 3, these effects can be minimized by taking sufficiently rapid measurements.

IV. Conclusions

This experiment shows how a simple scheme with a CW modulating signal can yield a measure of group delay, which can be used to derive the chromatic dispersion. The delay is not directly registered, instead a vector voltmeter is able to measure the phase difference $\Delta\phi$ between the original and the fiber routed signal.

However, the empirical results in Section III-A show that the phase shift for a given wavelength is not a constant value, but it slightly changes over time. Even if an explanation of the difference in the phase of the collected data was due in Section III-B, this apparent inconsistency does not affect the validity of the measure of the D-parameter, which is the main goal of the experiment, as explained before.

As anticipated in the introduction, the D-factor dependence on wavelength can be used to classify different types of fibers. The subdivisions is regulated by the ITU-T standards, under the series G.6xx. As regards single-mode fibers, the standards are summarized in Table II.

Denomination	Fiber type
G.652	Standard
G.653	Dispersion-shifted
G.654	Cut-off shifted
G.655	Non-zero dispersion-shifted

TABLE II: Single-mode fiber ITU-T standards.

In our case, the D-parameter is very low with respect to a standard fiber, for which it should be around 17 ps/nm-km. Hence, the fiber is certainly a dispersion-shifted one: in fact, a standard single-mode fiber has a zero dispersion wavelength around 1300 nm, whereas the measured one is around 1580 nm. However, the zero-dispersion wavelength of a dispersion-shifted fiber, as described in the standard G.653, should be 1500 nm. We deduce that the fiber may be a non-zero dispersion-shifted one.

In conclusion, the fiber under test is compliant with respect to the ITU-T G.655.C standard, as far as can be said with the experimental result, which is not complete because it lacks the response from 1530 nm to 1550 nm. Therefore, our measurements are compatible with a non-zero dispersion-shifted fiber.